

TrES-3b

R-2745

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210703 · created 2026-09-02T01:54:46+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459398.8993 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.174 +/- 0.055
Transit depth (Rp/Rs)^2	3.04 +/- 1.93 [%]
Orbital Inclination (inc)	78.97 +/- 2.93
Airmass coefficient 1 (a1)	2.44 +/- 0.52
Airmass coefficient 2 (a2)	-0.76 +/- 0.16
Scatter in the residuals of the lightcurve fit is	16.04 %
Best Comparison Star	#1 - [503, 385]
Optimal Aperture	3.32
Optimal Annulus	9.02
Transit Duration (day)	0.022 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.174 ± 0.055 vs 0.16309 (deviation 0.2σ)
Duration fit vs published	0.022 d vs 0.0567 d (deviation 1.28σ)
Mid-time O–C	-2.7 min
Depth SNR	3.2
Residual scatter	16.04 %

Light curve

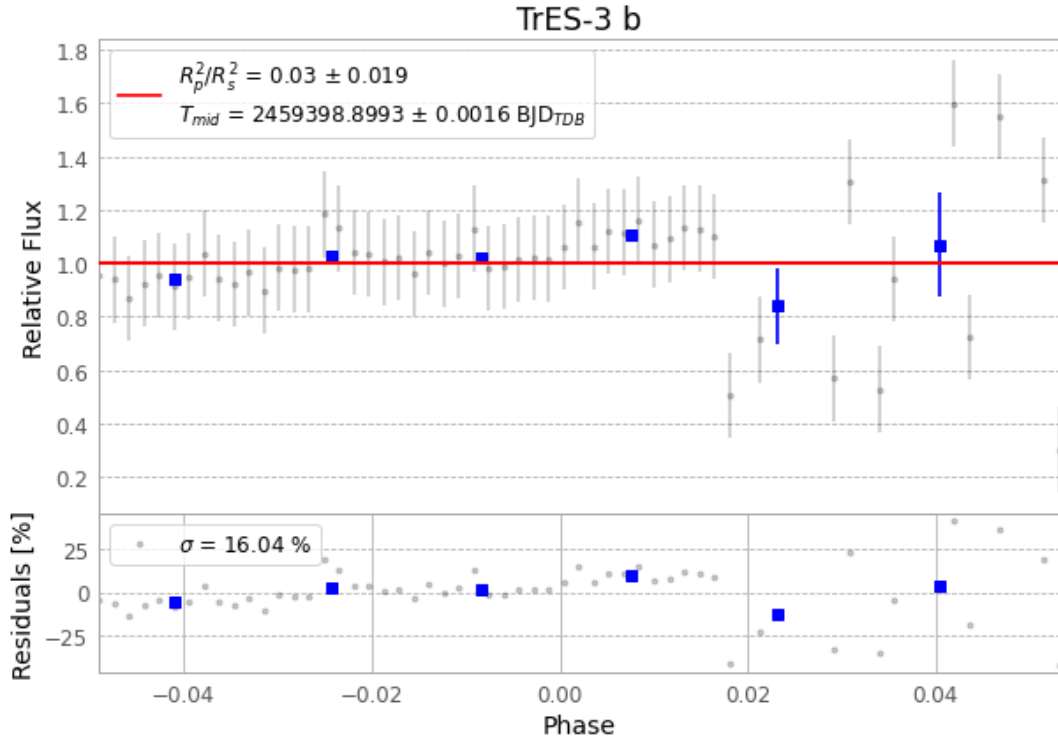


Figure 1 — FinalLightCurve_TrES-3 b_2021-07-03.png (SHA-256 1bbb151aea1dfe4d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [397, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 36f2e940ef7d67c9...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

67 FITS frames (44 MB), TRES-3210703075712.FITS → TRES-3210703111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210703.log (4a8c112e123e...), FinalLightCurve_TrES-3 b_2021-07-03.png (1bbb151aea1d...), FinalLightCurve_TrES-3 b_2021-07-03.csv (6cdb1089a85b...)

Compute resources

Wall time 272.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:54Z.